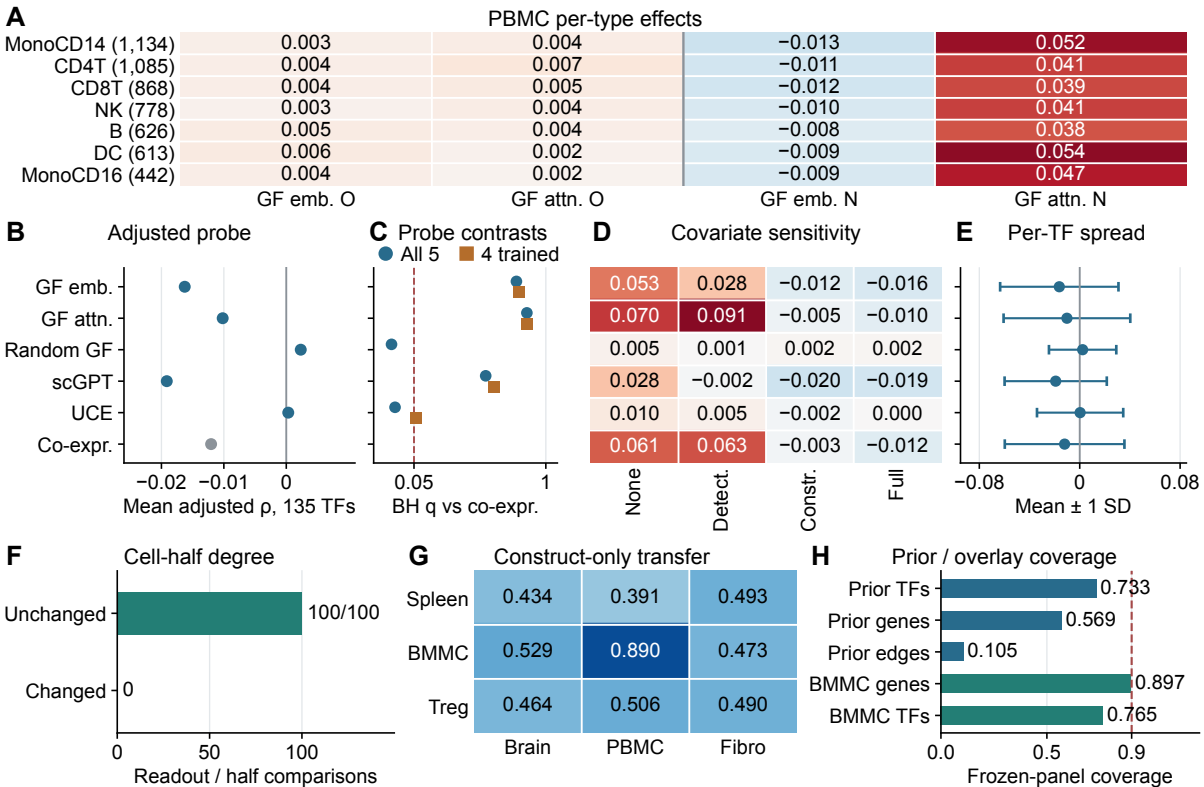
A
PBMC per-type effects
MonoCD14 (1,134) 0.003 0.004 −0.013 0.052
CD4T (1,085) 0.004 0.007 −0.011 0.041
CD8T (868) 0.004 0.005 −0.012 0.039
NK (778) 0.003 0.004 −0.010 0.041
B (626) 0.005 0.004 −0.008 0.038
DC (613) 0.006 0.002 −0.009 0.054
MonoCD16 (442) 0.004 0.002 −0.009 0.047
GF emb. O
GF attn. O
GF emb. N
GF attn. N
B
Adjusted probe
GF emb.
GF attn.
Random GF
scGPT
UCE
Co-expr.
−0.02 −0.01 0
Mean adjusted ρ, 135 TFs
C
Probe contrasts
All 5
4 trained
0.05 1
BH q vs co-expr.
D
Covariate sensitivity
0.053 0.028 −0.012 −0.016
0.070 0.091 −0.005 −0.010
0.005 0.001 0.002 0.002
0.028 −0.002 −0.020 −0.019
0.010 0.005 −0.002 0.000
0.061 0.063 −0.003 −0.012
None
Detect.
Constr.
Full
E
Per-TF spread
−0.08 0 0.08
Mean ± 1 SD
F
Cell-half degree
Unchanged
Changed
100/100
0
0 50 100
Readout / half comparisons
G
Construct-only transfer
Spleen 0.434 0.391 0.493
BMMC 0.529 0.890 0.473
Treg 0.464 0.506 0.490
Brain
PBMC
Fibro
H
Prior / overlay coverage
Prior TFs 0.733
Prior genes 0.569
Prior edges 0.105
BMMC genes 0.897
BMMC TFs 0.765
0.0 0.5 0.9
Frozen-panel coverage